Predictive Modelling of Breast Cancer Survival Outcome

EYRAUD – GOIGNARD

ENSIIE – MRR Project

December 15, 2025

Outline

Outline

Motivation & Objective

- Breast cancer is highly heterogeneous; survival prediction is crucial for prognosis.
- Goal: Identify **gene expression patterns** and **clinical factors** influencing survival.
- Binary outcome: **alive vs deceased**.
- Data:
  - 1,231 patients
  - 24 clinical variables
  - 5,000 gene expressions

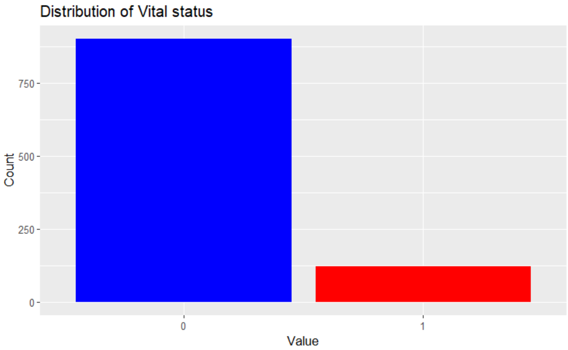
Outline

Data Pre-processing

- Converted all clinical non-numeric variables into factors.
- what we did :
  - grouped rare levels (too few observations),
  - removed variables with near-zero variance,
  - removed uninformative variables/observations (`gender`, `primary_site`)
- Final sample size after cleaning : **1022 patients**

Target Variable Distribution

$$\texttt{vital\_status} = \begin{cases} 0 & \text{if the patient is alive,} \\ 1 & \text{if the patient is deceased.} \end{cases}$$



The high-dimensionality problem

We have joined the clinical and Genes datasets on the patients id, it contains :

- 1022 observations
- but 5011 variables !

$\longrightarrow$ GLM models will not converge/be too large ! We need a work around

Gene Expression Clustering: PAM50 vs. K-means

- **Methodology**: Applied K-means ($k = 5$) to 5000 scaled gene expressions.
- **Validation**: Compared K-means clusters to established PAM50 subtypes.
- **Result**: Strong agreement on subtypes (Basal, Normal), with the main discrepancy in separating Luminal A and B and detecting HER2.

Table: K-means Cluster vs. PAM50 Subtype Contingency Table (Gene-level)

PAM50 Subtype	Cluster 1	Cluster 2	Cluster 3	Cluster 4	Cluster 5
Basal	198	1	3	7	1
HER2	15	50	83	0	1
LumB	0	262	138	0	0
LumA	0	94	220	7	19
Normal	0	1	15	13	103

The cleaned datasets

What we're left with

Two dataframes containing both clinical and genes expression data :

- The *complete* dataframe - 1022 obs. and **5011 variables**
- The *clustered* dataframe - 1022 obs. and **12 variables**

Outline

Train/Test Splitting and Imbalance

We tested the regression models on both datasets

- **Dataset split** : 80% training (and CV), 20% held-out test set
- **Outcome imbalance** : Only ∼**12%** of patients are deceased
- **Solutions tested** :
  - **Upsampling** the minority class (training only)
  - **Class-weighted** models on the original dataset
- **Evaluation** : Final performance assessed on the untouched test set (20% split).

Model Evaluation Strategy

- **Metrics** : Accuracy, **Recall**, Precision, **F1-score**.
- **Threshold Optimization**:
  - Due to class imbalance, the threshold $p = 0.5$ is often not optimal.
  - We optimized the threshold on a validation set (or cross-validation) to maximize the **F1-score**.

Outline

Simple Models : The Principle

Generalized Linear Model (Logistic Regression)

- Used to model the probability of death $p = P(\texttt{vital\_status} = 1)$.
- The relationship is defined by the log-odds (logit link function):

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k$$

$\longrightarrow$ This type of models are suitable only for low-dimensionality datasets e.g $p < n$

Simple models : Full & Variable Section Models

The full model :

- contains all of the 12 variables

The Variable Selection Models

- Build more parcimonious models
- Methods : Forward, Backward, and Stepwise (both)
- Selection Criteria :
  - **AIC** : Minimizes $\text{AIC} = -2 \cdot \log(\text{L}) + 2p$
  - **BIC** : Minimizes $\text{BIC} = -2 \cdot \log(\text{L}) + p \cdot \log(n)$

(L is the likelihood, p is the number of parameters, n is the sample size)

Advanced Models : The principle

Objective : Overcome the $p \gg n$ problem and improve stability by penalizing variables

Table: Comparison of penalized logistic regression models

Model	α	**Penalization Type**	**Feature Selection**
Ridge	0	L2	No
Lasso	1	L1	Yes
Elastic Net	$[0, 1]$	L1 and L2	Partial

Outline

Simple Regression Models

- Models based on simple Logistic Regression with variable selection (AIC/BIC criteria)
- Using the *clustered* dataset
- Performance evaluated on the 20% test set with 0.5 treshold

Table: Performance of the best simple Logistic Regression Models

Model	Accuracy	Recall	Precision	F1-score
Full Model	0.779	0.500	0.267	0.348
Stepwise-AIC	0.750	0.667	0.271	0.386
Stepwise-BIC	0.804	0.417	0.278	0.333

Advanced Regression Models - Ridge

Table: *K-clustered* vs *C-complete* and λ_{min} vs λ_{1SE}

Model	**Threshold** (λ_{min}, λ_{1SE})	**F1-score** (λ_{min}, λ_{1SE})	**Recall** (λ_{min}, λ_{1SE})
K-Ridge	(0.22, 0.16)	(0.367, 0.400)	(0.375, 0.333)
K-RidgeUp	(0.38, 0.39)	(0.260, 0.252)	(0.667, 0.667)
K-RidgeW	(0.57, 0.57)	(0.375, 0.367)	(0.500, 0.375)
C-Ridge	(0.10, 0.11)	(0.298, 0.280)	(0.583, 0.625)
C-RidgeUp	(0.32, 0.32)	(0.322, 0.315)	(0.583, 0.583)
C-RidgeW	(0.43, 0.46)	(0.280, 0.271)	(0.542, 0.542)

Advanced Regression Models - Lasso

Table: *K-clustered* vs *C-complete* and $\lambda_{\min}$ vs λ_{1SE}

Model	**Threshold** ($\lambda_{\min}$, λ_{1SE})	**F1-score** ($\lambda_{\min}$, λ_{1SE})	**Recall** ($\lambda_{\min}$, λ_{1SE})
K-Lasso	(0.14, 0.13)	(0.384, 0.313)	(0.583, 0.417)
K-LassoUp	(0.38, 0.37)	(0.267, 0.236)	(0.667, 0.625)
K-LassoW	(0.54, 0.51)	(0.394, 0.368)	(0.583, 0.583)
C-Lasso	(0.21, 0.10)	(0.327, 0.251)	(0.375, 0.958)
C-LassoUp	(0.18, 0.27)	(0.393, 0.407)	(0.500, 0.500)
C-LassoW	(0.59, 0.48)	(0.292, 0.259)	(0.292, 0.875)

Advanced Regression Models - Elastic Net

Table: *K-clustered* vs *C-complete* and $\lambda_{\min}$ vs λ_{1SE}

Model	α	**Threshold** ($\lambda_{\min}$, λ_{1SE})	**F1-score** ($\lambda_{\min}$, λ_{1SE})	**Recall** ($\lambda_{\min}$, λ_{1SE})
K-Enet	0.05	(0.14, 0.17)	(0.373, 0.400)	(0.583, 0.333)
K-EnetUp	0.15	(0.27, 0.28)	(0.316, 0.343)	(0.500, 0.500)
K-EnetW	0.05	(0.58, 0.52)	(0.381, 0.388)	(0.500, 0.542)
C-Enet	0.05	(0.13, 0.11)	(0.320, 0.275)	(0.500, 0.625)
C-EnetUp	0.10	(0.59, 0.57)	(0.386, 0.379)	(0.458, 0.458)
C-EnetW	0	(0.43, 0.45)	(0.286, 0.275)	(0.542, 0.583)

Key Takeaways : Data and Regularization

- **Raw Gene Data Boosts Prediction:** Models using all 5,000 genes (C-complete) achieved the highest F1 (0.407), showing that cluster summaries lose critical prognostic information
- **Penalty Choice Drives Objective:**
  - **Lasso (L1)** maximizes sensitivity: C-Lasso λ_{1SE} Recall $= 0.958 \rightarrow$ detects nearly all high-risk patients
  - **Ridge (L2)** stabilizes small datasets: K-Ridge λ_{1SE} F1 $= 0.400 \rightarrow$ robust performance on clustered features
- **Handling Imbalance is Critical:**
  - **Upsampling** hurts Precision $\rightarrow$ lowers F1 (e.g., K-LassoUp F1 ~ 0.26)
  - **Class weighting** balances Recall and F1, stabilizing predictions (e.g., K-RidgeW F1 ~ 0.375) while raising the optimal threshold

Advanced Regression Models – Accept–Reject Lasso (ARL)

- ARL uses **stability selection** to identify reliably selected variables
- Ensures an upper bound on false selections through the **PFER** parameter
- PFER $= 10$; cutoff $= 0.65$

Table: Performance of ARL-based Lasso Models

Model	#Features	Threshold	F1-score	Recall
ARL Lasso	19	0.18	0.24	0.33

Advanced Regression Models – Variable Decorrelation Lasso

- Highly correlated genes hurt Lasso stability
- Solution : cluster genes by correlation and extract the **first principal component** of each cluster
- Reduces dimensionality from 5000 genes to ∼50 uncorrelated components

Table: Performance of the Decorrelated Lasso Model

Model	**Threshold**	**F1-score**	**Recall**
Decorr-Lasso	0.19	0.4	0.42

Advanced Regression Models – UniLasso

- Filters genes via **univariate logistic regression**.
- Only genes with significant p-values (e.g. $p < 0.05$) are kept.
- Then a Lasso model is trained on this reduced, biologically relevant subset

Table: Performance of the UniLasso Model

Model	#Genes Selected	Threshold	F1-score	Recall
UniLasso	1878	0.13	0.3	0.5

Key Takeaways : Specialized Models and Performance

- **Best Balanced Performance (F1) :** C-EnetUp ($\alpha = 0.10$, λ_{1SE}) reached 0.407 F1 $\rightarrow$ best trade-off between sensitivity and precision
- **Highest Sensitivity (Recall) :** C-Lasso λ_{1SE} Recall $= 0.958 \rightarrow$ nearly all high-risk patients detected, but Precision is low ($\sim$ 25% correct positives)
- **Specialized Lasso Insights :**
  - **Decorrelation Lasso** : F1 $= 0.40 \rightarrow$ handling gene correlations stabilizes predictions
  - **UniLasso** : 60% feature reduction, F1 $= 0.30$, Recall $= 0.50 \rightarrow$ interpretable without huge performance loss
- **Threshold Tuning is Key :** $\sim 0.15 \rightarrow$ optimal F1; $\sim 0.10 \rightarrow$ maximize Recall (detect more high-risk patients)

Outline

Conclusion : Key Insights

Achievements

- Overcame the $p \gg n$ challenge in breast cancer survival prediction using penalized logistic regression
- The biggest problem remains the strong imbalance
- Including all 5000 gene expressions substantially improves predictive power over clinical data alone

Top Models

- **Balanced Performance :** C-EnetUp ($\alpha = 0.10$, λ_{1SE}) F1 $= 0.407 \rightarrow$ best trade-off between detecting high-risk patients and prediction reliability.
- **Maximum Sensitivity :** C-Lasso (λ_{1SE}) Recall $= 0.958 \rightarrow$ nearly all high-risk patients identified, essential for prognosis-focused decisions.

Thank you!